D 13611 **(Pages : 3)** **Name.....**

Reg. No.....

FIRST SEMESTER (CBCSS-UG) DEGREE EXAMINATION
NOVEMBER 2021

Physics/Applied Physics

PHY 1B 01/APH 1B 01—MECHANICS—I

(2020 Admissions)

Time : Two Hours Maximum : 60 Marks

The symbols used in the question paper have their usual meanings.

Section A (Short Answer Type)

Answer **all** *questions in two or three sentences.*
Each correct answer carries a maximum of 2 marks.

1. With proper examples define contact forces.
2. Explain Fictious or Pseudo force.
3. Explain the gravitational force of a sphere.
4. Draw the force diagram for the "Dangling rope" and find the tension a distance x from the bottom.
5. State the law of conservation of linear momentum with one example.
6. Define conservative and non-conservative forces. Give examples of each.
7. State and explain the work energy theorem.
8. Draw the energy diagram for a harmonic oscillator and explain.
9. Define power with its various units.
10. Explain torque and compare with force.
11. State and explain parallel axis theorem.
12. Sketch and deduce an equation for the period of a Simple Pendulum.

(Ceiling 20 marks)

Section B (Paragraph / Problem type)

Answer **all** *questions in a paragraph of about half a page to one page.*
Each correct answer carries a maximum of 5 marks.

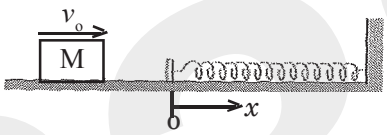
13. A string of mass m attached to a block of mass M is pulled with force F. Neglect gravity. What is the force F_1 on the block due to the string ?

Turn over

14. Mass m is whirled on the end of a string length R. The motion is in a vertical plane in the gravitational field of the earth. The forces on m are the weight W down, and the string force T towards the centre. The instantaneous speed is v, and the string makes angle θ with the horizontal. Find the T and the tangential acceleration at this instant.

15. A rod of length L has a non-uniform density. λ, the mass per unit length of the rod, varies as $\lambda = \lambda_0 (s / L)$, where λ_0 is a constant and s is the distance from the end marked 0. Find the centre of mass.

16. A block of mass M slides along a horizontal table with speed v_0. At $x = 0$, it hits a spring with spring constant k and begins to experience a friction force, as indicated in the sketch. The co-efficient of friction is variable and is given by $\mu = bx$, where b is a constant. Find the distance l the block travels before coming to rest :



17. A particle of mass m moves in one dimension along the positive x axis. It is acted on by a constant force directed toward the origin with magnitude B, and an inverse-square law repulsive force with magnitude A/x^2.

    (a) Find the potential energy function U (x).

    (b) Sketch the energy diagram for the system when the maximum kinetic energy is $K_0 = \frac{1}{2} m v_0^2$.

    (c) Find the equilibrium position, x_0.

18. A uniform drum of radius b and mass M rolls without slipping down a plane inclined at angle θ. The moment of inertia of the drum around its axis is $Io = Mb^2/2$. Find the drum's acceleration along the plane.

19. Find the moment of inertia of a thin sheet of mass M in the shape of an equilateral triangle around an axis through a vertex, perpendicular to the sheet. The length of each side is L.

(Ceiling 30 marks)

Section C (Essay Type)

Answer in about **two** *pages, any one question.*
Answer carries 10 marks.

20. State Newton's laws of motion. Apply Newtons laws to find the accelerations of two astronauts of masses M_A and M_B pulling on either ends of a rope of negligible mass.

21. State and explain law of conservation of Angular Momentum for a system of particles. Express the torque acting on a rigid body in a uniform gravitational field in terms of position of centre of mass and the weight of the body.

(1 × 10 = 10 marks)